

CELLCHECK · CALCULATION NOTE

CellCheck sizing calculations

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DOCUMENT CONTROL

Document ID	CCK-CAL-001	Version	0.5
Status	Draft	Date	2026-10-02
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/cellcheck		
Source	docs/04-calcs/01-sizing.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-02	Amish Chadha	R5 row of Table 4 names the same-day scheduling rule and section 8 the tray decision of 2026-10-02 (CCK-DEC-001, items 4 and 5); no number changed
0.4	2026-10-01	Amish Chadha	Budget treated as a value-engineering target; cost wording only, no number changed
0.3	2026-10-01	Amish Chadha	Design made constructable (CCK-DDR-003). Size and mass (section 8) and cost (section 11) rerun for the 1.5 mm base plate on six standoffs, rubber feet, folded guard and shroud, fixings and the 5 V converter
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). Budget \$175 (R12 met); temperature-gated resistance pulse (R4 met); same-channel, cohort-median self-discharge reading (R6)
0.1	2026-09-25	Amish Chadha	First TRL 3 sizing note (cycle time and throughput, power, heatsink, measurement error budgets, single-fault analysis, containment, mass,

grading on synthetic data, second-life
SwapCell study, cost)

CellCheck sizing calculations

On paper, CellCheck meets 13 of its 17 requirements, and no requirement is failed outright. Version 0.2 applies the decisions Amish accepted on 2026-09-25 (CCK-DDR-002): the value-engineering target in `project.yaml` is now \$175 (a hypothetical control target, not a limit), so the estimated parts cost is within the target for **R12** (it was \$4 over the former \$160 target), and the resistance pulse now starts only within 1 K of the bench reading, so **R4** is met at $\pm 2.09 \text{ m}\Omega$ against $\pm 3.0 \text{ m}\Omega$ (it was $\pm 2.99 \text{ m}\Omega$, at risk). Three requirements remain **at risk**: R5 (throughput is exactly 12 cells per attended day), R6 (self-discharge reading) and R11 (mass is 4.91 kg against 5 kg). R10 (containment) cannot be verified at TRL 3. The TRL 2 figures change in several places: time per cell falls from about 6.5 to 6.33 h, capacity error without calibration is $\pm 1.1 \%$ rather than $\pm 2 \%$, voltage error without calibration is $\pm 12.4 \text{ mV}$ rather than $\pm 10 \text{ mV}$, and the unit is 85 mm high and 4.91 kg rather than 90 mm and 3.5 kg. The heatsink fins in the TRL 2 massing model ran across the fan airflow; they are now vertical plates in line with it. Version 0.3 reruns size, mass and cost for the constructable design of CCK-DDR-003 (feet, a thinner base plate on six standoffs, fixings and a 5 V converter): 85 mm high, 4.91 kg and \$172.40, so R11 stays at risk and R12 stays met with \$2.60 of margin. No other result changes.

Every number here is printed by `docs/04-calcs/sizing.py` (run from the repo root: `python docs/04-calcs/sizing.py`), which also writes `docs/04-calcs/results.csv`. The script reads the geometry from `cad/src/model.py` and the costs from `bom/bom.csv`. All values are first-principles estimates; nothing is measured.

■ 1. Assumptions

Table 1. Inputs. All are assumptions for a paper design.

Input	Value	Basis
Typical cell	18650, 1.8 Ah as found, arriving at 30 % state of charge; also 2.5 Ah (healthy laptop cell) and 4.5 Ah (salvaged 21700)	Near the 1.62 Ah mean in Olivero-Ortiz et al. (2026), cited in CCK-PRB-001
Charge	1.0 A CC to 80 % state of charge, then CV at 4.20 V with an exponential decay to 0.1 A	Typical at about 0.5C
Recharge for the rest	1.0 A CC until the terminal reads 4.10 V under current, about 80 % state of charge	Charger modules are fixed at 4.20 V, so the firmware stops them
Discharge	1.0 A to 2.80 V; delivered capacity equals the as-found capacity	
Rests, pulse, handling	0.5 h and 0.25 h rests; 11 s resistance pulse; 0.5 h channel idle per cell for handling and 0.02 h for the day-14 reading	Handling allowance is conservative
Attended day	10 h; charge, pulse, discharge and handling need a person present; rests do not	CCK-DDR-001 item 2
Mean cell voltage	3.90 V on charge, 3.65 V on discharge, 4.20 V worst case at the start of discharge	
Efficiencies	Charger module 85 %; adapter 88 %; controller, display and fans 3 W	
Monitor (INA226 class)	Shunt offset 10 μ V; gain error 0.1 %; bus offset 7.5 mV; bus gain error 0.1 %; bus resolution 1.25 mV	Typical data sheet maximums; to be confirmed against the chosen part
Shunt	25 m Ω , 0.5 %, 100 ppm/K over a 20 K swing	
Calibration	Two-point voltage calibration per channel against a meter of 0.05 % + 2 counts (1 mV); current gain calibrated to 0.1 %	One-time, at build
Cell behaviour	Near the 2.80 V cut-off, 1 % of capacity per 50 mV; resistance changes 1.5 %/K; open-circuit voltage changes 0.2 mV/K	Assumptions for a generic lithium-ion cell
Resistance pulse start	Only when the cell reads within 1 K of the bench thermistor, so reinsertions differ by 1 K at most	Firmware rule, CCK-DDR-002 item 13
Temperature sensing	10 k Ω NTC, B 3950, 1 %, 10 k Ω divider from 3.3 V; ADC error 20 mV after factory calibration; 0.3 K for part tolerance	ESP32-class ADC; assumption
Heatsink	40 aluminium fins, 1.6 x 30 x 50 mm at 8 mm pitch on a 320 x 6 x 50 mm spine; case to sink 1.0 K/W through an insulating pad; junction to case 1.0 K/W	<code>cad/src/model.py</code>
Fans	60 mm, 8 L/s free air each, half of that through the fins; Nusselt number 10 in the fin channels; 5 W/(m ² K) effective with the fans stopped	Assumptions
Materials	Steel 7,850 kg/m ³ ; aluminium 2,700 kg/m ³ ; ceramic fibre sheet 128 kg/m ³	
Bought parts	Masses in <code>sizing.py</code> section 8 (holders 35 g each, modules 8 to 12 g, wiring 200 g)	Assumptions

■ 2. Cycle time and throughput (R5)

A typical 1.8 Ah cell occupies a channel for **6.33 h**: charge 0.90 h CC plus 0.92 h CV, rest 0.5 h, pulse, discharge 1.80 h, rest 0.25 h, recharge 1.44 h and 0.52 h of handling. A 2.5 Ah cell takes 8.30 h and a 4.5 Ah 21700 takes 13.93 h. The TRL 2 figure of about 6.5 h underestimated the CV phase and overestimated the recharge.

Because charge and discharge must be attended (CCK-DDR-001 item 2), throughput depends on the scheduling rule. A day-by-day simulation of the eight channels gives:

Table 2. Cells finished per day, steady state, eight channels.

Rule	1.8 Ah	2.5 Ah	4.5 Ah
A stage with current starts only if it finishes within the 10 h day	12.0	8.0	4.8
Current stages pause overnight with the unit switched off, and resume	12.8	9.6	5.6
Reference only: running 24 h, handling in attended hours (not allowed)	16.0	16.0	8.0

On the first rule the grader finishes exactly 12 typical cells a day, so **R5 is at risk** with no margin, and it is not met on healthier 2.5 Ah cells. The pause rule gains 0.8 cells a day, but a discharge interrupted overnight lets the cell recover and adds a small capacity error that is not quantified here. One 4S6P pack needs about 36.4 candidates at the 66 % yield assumed in the flow diagram, so about 3.0 attended days plus the 14-day rest.

■ 3. Power and energy (R13)

With all eight channels charging at 4.2 V and 1 A, the adapter supplies **42.5 W** (71 % of 60 W), or 3.54 A at 12 V; the 6.3 A main fuse is 1.78 times that. With the adapter at +5 % the highest voltage inside the unit is **12.6 V**, under the 13 V of R13.

Each typical cell takes 4.91 Wh on its first charge and 5.62 Wh on its recharge, plus 2.38 Wh of overhead, for **16.8 Wh** from the mains (about \$0.0025 at \$0.15/kWh). The load burns 6.57 Wh per cell as heat.

■ 4. Heatsink and MOSFET temperature (R8)

At full discharge load the eight MOSFETs dissipate **29.2 W** on average and **33.6 W** at the start (4.2 V x 1 A x 8). To keep the MOSFET cases at 75 °C in a 35 °C room, the heatsink must reach **1.07 K/W** or better.

The fans push air into the back of the fin channels through a shroud, and the air leaves through the open tops. With 8.0 L/s through 39 channels the air speed is 0.64 m/s, the heat transfer coefficient is 20.3 W/(m² K) on 0.120 m² of fin area, and the fin efficiency is 0.964. Adding half of the 3.6 K air temperature rise and 0.10 K/W of spreading in the spine gives **0.58 K/W**. The MOSFET cases then reach **58.7 °C** and the junctions 62.9 °C, so **R8 is met**. If both fans stop, the heatsink falls to about 1.78 K/W and the cases would settle near **99 °C**, so the firmware must stop all discharges when the fan tachometer or a heatsink thermistor reports a fault. The 2.0 A resistance pulse adds 8.4 W to one MOSFET for 1 s, which is small against its thermal mass.

Cell heating at 1 A is small: 0.06 W and about 1.4 K for a 60 mΩ cell, 3.6 K at the 150 mΩ grade C limit and 10.0 K for the worst 418 mΩ cell reported by Olivero-Ortiz et al. (2026). The 8 K rise limit therefore flags high-resistance cells without tripping on healthy ones.

■ 5. Measurement error budgets (R2, R3, R4, R6, R7)

Capacity (R2). Without calibration the worst case is the sum of the shunt tolerance (0.5 %), monitor gain (0.1 %), current offset (0.04 % at 1 A), shunt temperature drift (0.2 %) and the cut-off voltage error (12.4 mV, worth 0.25 % of capacity): **± 1.09 %**. After calibration it is **± 0.44 %**. R2 (± 2 %) is met.

Voltage (R3). Without calibration the worst case is 7.5 mV offset, 4.3 mV gain error at 4.3 V and half a count: **± 12.4 mV**, which does not meet ± 10 mV. After a two-point calibration of each channel against a 0.05 % meter it is **± 4.8 mV**. R3 is met only with the per-channel calibration, which is now part of the design.

DC resistance (R4). A 1.25 mV count over the 1.5 A current step gives **0.83 m Ω per count**. For a 60 m Ω cell, the repeatability over reinsertions sums to **± 2.09 m Ω** worst case (0.83 m Ω quantization, 0.36 m Ω gain, 0.90 m Ω from a 1 K cell temperature difference) and ± 1.28 m Ω RSS, against a limit of ± 3.0 m Ω . Offsets cancel in the difference, and the four-wire contacts keep contact resistance out. The firmware starts the pulse only when the cell is within 1 K of the bench reading (CCK-DDR-002 item 13); version 0.1 allowed 2 K and found ± 2.99 m Ω , at risk. **R4 is met** on the error budget. The 30 min rest before the pulse is normally enough for a cell that warmed by about 1.4 K on charge; a warmer cell waits longer, which this note does not add to the cycle time.

Self-discharge (R6). If the day-14 reading is taken in a different channel from the day-0 reading, the two calibrated readings and a 5 K room change give **± 10.6 mV** worst case (± 6.8 mV RSS) against ± 3 mV. In the same channel, offset and gain cancel and the error falls to **± 2.3 mV**. In addition, a cell whose charge stops under current relaxes by an amount this note cannot bound. Under CCK-DDR-002 item 13 each cell now returns to the channel it was graded in, so the reading error is **± 2.3 mV**, inside ± 3 mV, and its drop is judged against the median drop of the cells charged on the same day, which cancels the common part of relaxation and room temperature. The spread of relaxation between cells is still not bounded on paper, so **R6 remains at risk**.

Temperature (R7). The divider gives 27.2 mV/K at 45 °C; a 20 mV ADC error plus 0.3 K of part tolerance is **± 1.04 K**, inside ± 2 K. Sampling at 1 Hz is a firmware setting. The thermal lag of the clip is not analysed and is not verifiable at TRL 3.

■ 6. Single-fault analysis (R9)

This is the failure mode and effects analysis called for at TRL 2, with the hardware watchdog and undervoltage comparators adopted in CCK-DDR-001 item 3. The watchdog removes every charge enable and every load gate drive when the controller stops refreshing it. Each channel's comparator removes that channel's load gate drive below 2.5 V. Each cell has a 3 A fast fuse in its positive lead, and the charge switch is in series between the charger and the cell.

Table 3. Single faults against the R9 limits (above 4.25 V, above 60 °C, below 2.5 V).

Single fault	Effect without protection	What stops it	Result
Controller crash or firmware hang	Charger and load stay as last set; temperature limits lost	Watchdog opens all charge switches and removes all load gate drive	Safe
Load MOSFET fails short	Cell discharges through the channel	Short current is 31.3 A for a 60 mΩ cell and at least 5.9 A for a 418 mΩ cell at 2.8 V, 2.0 times the 3 A fuse or more; the fuse opens	Safe; cell rejected
Op-amp or gate drive stuck on	Load runs past the cut-off	Undervoltage comparator removes the gate drive at 2.5 V	Safe
Charger output regulates high (failed feedback)	Cell charged above 4.20 V	Firmware reads the monitor and opens the series charge switch at 4.25 V; current above 1.2 A also opens it	Safe
Charge switch fails short	Charger stays connected	Charger module limits the cell to 4.242 V (4.20 V + 1 %)	Safe
Thermistor open or shorted	Temperature reads out of range	Firmware treats a reading outside -20 to 80 °C as a fault and stops the channel	Safe
Thermistor clip detached	Reads the air, not the cell	Not detected; charger voltage limit still holds	Residual risk
Fan stops	MOSFETs could reach about 99 °C	Firmware stops discharges on a tachometer or heatsink fault	Safe (firmware)
Adapter output high	Bus voltage rises	Certified adapter; charger modules regulate their own output	Safe for R9; R13 depends on the adapter

R9 is met on analysis for the faults it lists. The detached thermistor is a residual risk that R9 does not name; a plausibility check (cell temperature must rise slightly during discharge) narrows it and is now part of the firmware rules (CCK-DDR-002 item 13), but it is not verified at TRL 3. The fuses and comparators are not verified by test at TRL 3.

■ 7. Containment (R10)

The 1.5 mm steel tray has 0.206 m² of sheet and weighs **2.43 kg**. If a 21700 cell in thermal runaway released 100 kJ in total, including burning vent gas (an assumption that must be replaced by a sourced figure before TRL 4), the mean tray temperature would rise about **84 K** if the tray absorbed all of it. Mean temperature says little about jets, hot spots on the bench under the tray, or flame through the perforated guard, and none of these can be analysed credibly on paper. **R10 is not verifiable at TRL 3**. The fan intake slots in the back wall (56 x 22 mm each) are a new path out of the tray and are noted for the containment review.

■ 8. Size and mass (R11)

The model envelope is **460 x 300 x 85 mm** (the display stand and the 8 mm rubber feet set the height), inside 500 x 320 x 120 mm. The mass without the adapter is **4.91 kg**: tray 2.43 kg, base plate 0.47 kg, heatsink 0.52 kg, guard 0.24 kg, shroud 0.06 kg, liner 0.05 kg, and 1.14 kg of bought parts, fixings and wiring. **R11 is at risk** with 0.09 kg of

margin. In version 0.2 the unit was 75 mm and 4.97 kg; the constructable design (CCK-DDR-003) adds feet, fixings, guard flanges and a 5 V converter (about 0.12 kg) and takes 0.16 kg off by making the base plate 1.5 mm thick on six standoffs instead of 2 mm on four. With six supports the plate spans at most 165 mm between standoffs, so it stays stiff. A 1.2 mm tray would save about 0.49 kg, but the tray is also the containment, so Amish decided on 2026-10-02 to keep the 1.5 mm tray and accept 4.91 kg, with the prototype weighed at TRL 4 (CCK-DEC-001, item 5).

■ 9. Grading and matching on synthetic data (R15)

The grading rules (CCK-PRC-001 Table 2) and the matching method were run on 150 synthetic cells drawn from a seeded random generator: capacity normal with mean 1.62 Ah and standard deviation 0.53 Ah (clipped to 0.04 to 2.50 Ah, the figures reported by Olivero-Ortiz et al., 2026), resistance log-normal with a 75 mΩ median (an assumption), and 2.2 Ah rated capacity. They produce 15 grade A, 47 grade B, 55 grade C and 33 rejected cells. For a 4S6P pack from grade B, the method picks the 24 cells with the narrowest capacity window, distributes them with a serpentine sort and then swaps cells between groups while that reduces the spread. The four parallel groups come out at 9.677 to 9.678 Ah, a **0.006 %** maximum deviation from the mean. R15 is met on synthetic data; real cells will be less tidy.

■ 10. Second-life SwapCell variant (study)

CCK-DDR-001 item 1 keeps SwapCell in the pitch and asks for a second-life variant to be studied. A 13S4P pack of grade A laptop cells at 80 % of 2.2 Ah holds **7.04 Ah and about 334 Wh**, and at the 0.5C limit for grade A it can deliver **3.52 A**, about **165 W**. That covers PowerBox's reference evening of 224.8 Wh from the pack 1.34 times (PBX-CAL-001), but not its 300 W inverter, which draws about 8.7 A at 39 V. An assumed 330 x 84 x 74 mm cell space inside the SwapCell body holds up to 80 cells of 18650 size, so 52 cells fit before the BMS and wiring are placed. The study therefore supports a low-current storage variant for PowerBox with the AC output limited to about 150 W, or a DC-only PowerBox, and rules out e-bike use. Any variant needs the SwapCell project's agreement and its own pack design; it is not a CellCheck deliverable.

■ 11. Cost (R12)

The 21 BOM lines total **\$172.40**, \$2.60 under the \$175 value-engineering target in `project.yaml`, which is a hypothetical control target, not a limit. Amish accepted the \$175 target on 2026-09-25 (CCK-DDR-002 item 4). **R12 is within the value-engineering target.** The increase over TRL 2 is line 18 (watchdog and comparators, \$5.00); the increase over version 0.2 (\$164.00) is lines 19 to 21, added for construction (rubber feet \$2.40, fixings \$4.00, 5 V converter \$2.00).

■ 12. Results against requirements

Table 4. Requirement status (at risk first; no requirement is failed outright). Values from `results.csv`.

ID	Value	Target	Status
R5	12.0 cells per day under the same-day rule decided on 2026-10-02 (CCK-DEC-001, item 4); 12.8 with an overnight pause, not used; 8.0 for 2.5 Ah cells	12 cells per attended day on 1.8 Ah cells	At risk
R6	± 2.3 mV in the same channel, judged against the same-day cohort median (± 10.6 mV across channels, no longer allowed); spread of relaxation not bounded	± 3 mV; flag above 50 mV	At risk
R11	460 x 300 x 85 mm; 4.91 kg	500 x 320 x 120 mm; 5 kg	At risk
R10	Mean tray rise 84 K for an assumed 100 kJ event; jets and flame not analysable	Hold one venting 21700 in the tray	Not verifiable at TRL 3
R1	Eight channels; holders for 18650 and 21700 cells up to 70 mm long	18650 and 21700; 8 channels or more	Met (design review)
R2	± 1.1 % uncalibrated; ± 0.4 % calibrated	± 2 %	Met
R3	± 12.4 mV uncalibrated; ± 4.8 mV after per-channel calibration	± 10 mV	Met (with per-channel calibration)
R4	0.83 m Ω per count; ± 2.09 m Ω worst case, ± 1.28 m Ω RSS at 60 m Ω , pulse gated within 1 K	1 m Ω ; ± 3 m Ω or ± 5 %	Met
R7	± 1.0 K at 45 °C; 1 Hz sampling	1 Hz, ± 2 K; stop at 45 °C or +8 K	Met (accuracy); response time not verifiable at TRL 3
R8	58.7 °C case with fans (0.58 K/W); 99 °C if the fans stop	75 °C or less at 35 °C	Met
R9	Watchdog, undervoltage comparators, 3 A fuses; detached thermistor narrowed by a plausibility check, still residual	No single fault past 4.25 V, 60 °C or 2.5 V	Met (analysis)
R12	\$172.40; \$2.60 under \$175	\$175 value-engineering target (<code>project.yaml</code>)	Within the value-engineering target (\$2.60 under)
R13	Certified 12 V adapter; 12.6 V highest; 42.5 W of 60 W	Certified 12 V; 13 V or less	Met (design review)
R14	Rest stage unattended; current stages attended	As redefined in CCK-REQ-001 v0.3	Met (design review)
R15	0.006 % group deviation on synthetic 4S6P	± 1 %	Met (synthetic data)
R16	CSV per cell; group and reject lots map to the ReflowEconomy passport v0.2 with two gaps (section 13)	CSV per cell; local export	Met (design review)
R17	Modules and through-hole parts; open licences	Buildable and open	Met (design review)

■ 13. Record mapping to the ReflowEconomy passport (R16)

CCK-DDR-001 item 9 (decided by Amish, CCK-DDR-002) maps the CellCheck record to ReflowEconomy's material passport schema v0.2. The per-cell CSV stays the primary record; one passport is issued per lot that leaves the bench.

Table 5. Passport fields for a CellCheck lot.

Passport field	Matched group of graded cells	Lot of rejected cells
batch_id	Group ID, for example CCK-G-0007	Reject lot ID
product_form	No suitable value (gap 1)	export-lot
parent_batch_ids	Intake batch IDs of the source packs	Same
material	Cell format and chemistry, for example Li-ion 18650 NMC	Same
grade	CellCheck grade A, B or C	reject
mass_kg	Cell count x measured or table mass	Same
origin	Collection point and country of the source packs	Same
identification.method	No suitable value (gap 2)	Same
processed_by , process , date	Workshop, cellcheck-grade , grading date	Same

Gap 1: the schema has no `product_form` for graded cells. Gap 2: `identification.method` does not list an electrical test. Both are raised in `docs/REVIEW.md` for the ReflowEconomy project; this repo does not edit that schema.

■ 14. Checks against earlier documents

Table 6. TRL 2 figures corrected by this note.

Quantity	TRL 2 (CCK-PRC-001 v0.2)	This note	Where fixed
Time per cell	about 6.5 h	6.33 h (typical); 8.30 h for 2.5 Ah	PRC v0.3, README
Throughput	about 12 per attended day	12.0 (stage-boundary rule), at risk	PRC v0.3, REQ v0.3, README
Capacity error uncalibrated	about ± 2 %	± 1.1 %	PRC v0.3, REQ v0.3
Voltage error uncalibrated	about ± 10 mV	± 12.4 mV; ± 4.8 mV calibrated	PRC v0.3, REQ v0.3
Heatsink needed and expected	1.1 K/W; 0.5 to 0.8 K/W	1.07 K/W; 0.58 K/W	PRC v0.3
Heatsink fins	Plates across the fan airflow	Vertical plates in line with it	Model, PRC v0.3
Energy per cell	about 18 Wh	16.8 Wh	PRC v0.3
Cell heating at 1 A	about 0.1 W, a few kelvin	0.06 W and 1.4 K at 60 m Ω	PRC v0.3
Height and mass	90 mm; about 3.5 kg (tray 2.0 kg)	85 mm; 4.91 kg (tray 2.43 kg); 75 mm and 4.97 kg in v0.2	PRC v0.5, REQ v0.5, README
Parts cost	about \$159	\$172.40; \$164.00 in v0.2	PRC v0.5, REQ v0.5, README, BOM notes
Rest voltage	"rest at 4.10 V"	Recharge stops at 4.10 V under 1 A, about 80 % state of charge	PRC v0.3

Safety: These are paper calculations for a device that charges and discharges salvaged lithium-ion cells. The single-fault analysis and the containment estimate are not evidence that the unit is safe. Operate it only attended during charge and discharge, on a non-flammable surface with a smoke alarm and an extinguisher or sand bucket within reach. CellCheck is a research and prototype tool, and its grades do not certify any cell or pack as safe.